TASK-6

INPUT

```
#include <stdio.h>

int main() {
    double balance, rate, repayment;
    double interest, totalInterest = 0;
    double payment;
    int month = 0;

    printf("Enter loan amount: ");
    scanf("%lf", &balance);

    printf("Enter monthly interest rate (%%): ");
    scanf("%lf", &rate);

    printf("Enter monthly repayment: ");
    scanf("%lf", &repayment);

    if (balance <= 0 || rate < 0 || repayment <= 0) {
        printf("Invalid input!\n");
        return 0;
    }

    rate = rate / 100.0;

    if (repayment <= balance * rate) {
        printf("Loan cannot be fully repaid.\n");
        return 0;
    }

    printf("\nMonth\tInterest\tPayment\t\tBalance\n");

    while (balance > 0.000001) {
        month++;

        interest = balance * rate;
        totalInterest += interest;
        balance += interest;

        if (repayment >= balance)
            payment = balance;
        else
            payment = repayment;

        balance -= payment;
```

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        if (balance < 0.000001)
            balance = 0;

        printf("%d\t%.2f\t\t%.2f\t\t%.2f\n",
               month, interest, payment, balance);

        if (month >= 360 && balance > 0) {
            printf("Safety limit reached.\n");
            return 0;
        }
    }

    printf("\nLoan fully repaid!\n");
    printf("Total months: %d\n", month);
    printf("Total interest: Rs %.2f\n", totalInterest);

    return 0;
}
```

Output

